longitudinal_health.csv: Dataset Description

A column-by-column description of the uploaded longitudinal health dataset.

Format note: every line in the file is wrapped in an extra pair of double quotes, so a normal CSV reader sees it as a single column. The quotes must be stripped before parsing. Everything below comes from the cleaned version.

Overview

- **Size:** 90,000 rows × 37 columns
- **Structure:** 3,000 users (`user_id` 1–3000), each with exactly 30 consecutive daily records, from 2024-01-01 to 2024-01-30.
- **Type:** a longitudinal (panel) dataset of wearable and health-tracker data. It looks synthetic, since the groups are uniform and the relationships are very clean.
- **Missing data:** only 6 columns have blanks, and the blanks are intentional. Each has a matching `*_measured` / `*_available` flag that agrees 100% with them.

Columns

1. Person-level attributes

These are constant for each user across all 30 days, except `weight_kg`, which drifts slightly.

Column	Meaning	Values
`user_id`	Unique person ID	1–3000
`age`	Age in years	18–79 (mean 48.6)
`sex`	Sex, coded 0/1	0 = 1,514 users, 1 = 1,486 users (the file doesn't say which is which)
`bmi`	Body mass index	18.0–41.5 (mean 26.1)
`smoking_status`	Smoker flag	0 = non-smoker (2,412 users), 1 = smoker (588)
`family_history_cvd`	Family history of cardiovascular disease	0 = no (2,107), 1 = yes (893)

Column	Meaning	Values
`fitness_level`	Fitness category	0 = low (1,183), 1 = medium (1,225), 2 = high (592)
`height_m`	Height in metres	1.45–2.00 (mean 1.70)
`weight_kg`	Body weight in kg	45–134 (mean 76.2). It changes day to day, by about 0.22 kg on average over the month.

2. Date

Column	Meaning	Values
`date`	Calendar date	2024-01-01 to 2024-01-30
`day_index`	Day number within the study	1–30

3. Activity

Column	Meaning	Values
`activity_type`	Main activity of the day	Rest (15,665), Walking (12,579), Mixed_Cardio (12,529), Yoga (12,415), Cycling (12,329), Strength (12,272), Running (12,211)
`steps`	Daily step count	0–19,742 (mean 6,611)
`heart_points`	Google Fit-style heart points for intensity	0–120 (mean 42)
`move_minutes`	Active minutes	0–141 (mean 42.5)
`calories_burned`	Total daily energy expenditure	1,133–2,627 kcal (mean 1,864)
`distance_km`	Distance covered	0–14.8 (mean 5.0)
`step_cadence`	Steps per minute	0–120 (mean 75.5)
`cycling_power_watts`	Cycling power output	0–346.5. It is non-zero only on Cycling days (avg about 150 W) and 0 for every other activity.

Activity types differ clearly. Rest days average about 1,800 steps and 11 active minutes, while Running days average about 10,900 steps and 70 minutes.

4. Heart and vitals

Column	Meaning	Values
`avg_heart_rate`	Average heart rate (bpm)	45–121 (mean 76.8)
`resting_hr`	Resting heart rate (bpm)	45–100 (mean 64.5)
`hrv`	Heart rate variability (ms)	10–118 (mean 53.4)
`bp_systolic`	Systolic blood pressure (mmHg)	90–200 (mean 133.5), 60% missing
`bp_diastolic`	Diastolic blood pressure (mmHg)	56–139 (mean 86.3), 60% missing
`spo2`	Blood oxygen saturation (%)	91–100 (mean 95.5)
`body_temp_c`	Body temperature (°C)	35.8–38.3 (mean 36.65)

5. Glucose

Column	Meaning	Values
`fasting_glucose`	Fasting blood glucose (mg/dL)	70–182 (mean 103.4), 75% missing
`postprandial_glucose`	Glucose after a meal (mg/dL)	86–226 (mean 140.4), 75% missing

6. Sleep and wellbeing

Column	Meaning	Values
`sleep_hours`	Hours slept	3–10 (mean 7.0), 25% missing
`sleep_efficiency`	Fraction of time in bed actually asleep	0.50–0.98 (mean 0.85), 25% missing
`fatigue_score`	Self-reported fatigue	0–10 (mean 4.1)

7. Nutrition and hydration

Column	Meaning	Values
`calories_consumed`	Daily calorie intake	1,200–3,840 (mean 2,462)
`water_intake_l`	Water intake in litres	0.5–5.0 (mean 2.3)

8. Measurement flags

These are True/False columns that say whether the matching readings exist on that day.

Column	Meaning	True / False
`sleep_data_available`	Sleep data recorded	67,540 / 22,460
`bp_measured`	Blood pressure recorded	35,991 / 54,009
`glucose_measured`	Glucose recorded	22,314 / 67,686

9. Target label

Column	Meaning	Values
`cardiometabolic_risk_state`	Daily cardiometabolic risk level, 0 (lowest) to 4 (highest)	0: 791, 1: 27,037, 2: 47,142, 3: 6,547, 4: 8,483

This looks like the prediction target. It changes over time for most users (only 14 of 3,000 hold a single value all month), and the last-day distribution shifts toward the higher-risk states.

Patterns in the data

Risk state	Resting HR	HRV	Systolic BP	Fasting glucose	Steps/day
0	52.5	82.4	112	91	10,154
1	57.4	66.7	123	97	8,365
2	63.9	52.6	133	102	6,597
3	69.5	43.0	144	110	5,283
4	87.1	20.5	162	131	1,793

- Higher fitness levels go with lower resting heart rate, higher HRV and more steps. The resting HR averages are 70, 63 and 56 bpm for fitness levels 0, 1 and 2.
- Smokers have slightly lower SpO_2 (95.0% vs 95.6%).

Things to watch for

- **Quoted lines:** fix the wrapping quotes before loading, or the file will parse as one column.
- **Missing values:** use the flag columns rather than imputing blindly. Blood pressure and glucose are mostly absent by design.
- **Repeated person-level columns:** columns such as age and BMI repeat 30 times per user, so split by `user_id` when you train and test to avoid leakage.
- **Unlabelled codes:** the 0/1 coding of `sex` and the meaning of the fitness and risk levels are inferred, not documented in the file.